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# Coupled Thermosphere–Ionosphere Dynamics During the 17 January 2013 Storm: GITM–TIDAS Comparison With Millstone Hill ISR Measurements

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**Key Points:**

- Meridional-wind-driven uplift supports initial SED enhancement, while reduced $O/N_2$ and downward ion motion contribute to the subsequent broad Ne depletion
- Global Ionosphere Thermosphere Model (GITM) and TIDAS show similar large-scale responses but differ in their representation of mesoscale wave structures and nighttime depletion
- GITM reproduces equatorward-moving traveling ionospheric disturbance (TID) structures observed by incoherent scatter radar (ISR), whereas the mesoscale features are smoothed in the TIDAS reconstruction

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**Abstract** This article analyzes a weak geomagnetic storm on January 17–18, 2013, using a first-principles model (Global Ionosphere Thermosphere Model (GITM) with an empirical assimilative ionospheric model (TIDAS), evaluated against observations in the form of Millstone Hill (MH) incoherent scatter radar (ISR) altitude profiles and global navigation satellite system (GNSS) derived keograms of ionospheric total electron content. Results show that the midlatitude ionosphere exhibits a clear two-phase evolution. From about 16 to 20 UT, both systems show an enhancement that develops into an SED feature and migrates equatorward. In GITM, the spatial correspondence among equatorward meridional winds, upward ion motion, and elevated $hmF_2$ indicates that meridional-wind-driven uplift is the primary mechanism associated with the initial enhancement. After about 22 UT, a depletion develops concurrently with a regional decrease and equatorward progression of $O/N_2$, indicating increasing composition control. Global Ionosphere Thermosphere Model and TIDAS agree on the large-scale morphology, including storm enhanced density (SED) and mid-latitude trough features, but differ in detail. Global Ionosphere Thermosphere Model tends to place features slightly west of TIDAS, overestimates wave amplitudes in the 265° to 270°E sector while matching better in the 285° to 290°E sector, and produces nightside electron densities that are too low due to missing plasmaspheric down-welling. TIDAS reproduces storm morphology but smooths mesoscale structure. Millstone Hill ISR profiles of ionospheric plasma parameters confirm southward propagating wave-like features between about 02 and 09 UT on January 18th. While GITM captures the timing and vertical reach of these features, they appear more muted when using TIDAS. These results reveal a wind-drift-composition physical pathway in which meridional-wind-driven uplift supports the initial SED enhancement, reduced $O/N_2$ and downward ion motion contribute to the later broad depletion, and localized wind variations produce mesoscale structuring. This warrants two improvements for better upper atmospheric modeling: the inclusion of a plasmaspheric exchange in physics-based models to correct nightside bias; and data assimilation settings that govern localized depletions and traveling ionospheric disturbances to better preserve composition- and neutral wind-linked imprints.

**Plain Language Summary** Geomagnetic storms from the Sun can disturb the thin layer of charged gas high above Earth known as the ionosphere. These disturbances can bend or delay radio waves that support GNSS based navigation, communications, and satellite operations. We studied a weak geomagnetic storm on January 17–18, 2013 using two tools. One is a physics model called GITM that simulates how the upper atmosphere and ionosphere evolve. The other is TIDAS, which reconstructs electron density from many measurements, including GNSS signals and radar. In both tools, the storm-time response begins with a daytime increase in electron density that develops into an equatorward-moving enhancement. The physics-based model shows that equatorward neutral winds lift the ionized gas to higher altitudes, where it is lost more slowly, helping to form and maintain this enhanced-density feature. Later that evening, the density dropped as the mix of neutral gases changed. In particular, the amount of atomic oxygen relative to molecular nitrogen decreased, which increased the loss of electrons. Results are evaluated against detailed observations from both the MH ionospheric radar and maps of total electron content from GNSS receivers. The radar shows wave-like patterns that travel and descend with time. The model reproduces the timing and shape of these waves but overestimates them in some places and is too weak at night. TIDAS matches the broad patterns but smooths out the small waves. These findings point to next steps for improving model based representation of ionospheric responses to the geomagnetic storm: include exchange with the plasmasphere in models and tune data methods so they retain small but important wind-driven structures.

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## 1. Introduction

Geomagnetic activity perturbs the Earth's ionosphere, leading to a complex array of plasma reconfigurations characterized by large-scale enhancements and depletions in electron density, restructuring neutral temperatures, winds, and plasma transport from the auroral zones to midlatitudes (Gonzalez et al., 1994; Buonsanto, 1999; Mendillo, 2006; J. C. Foster, 1993; Kil et al., 2003; Balan et al., 2011; Tsagouri et al., 2022; Upadhyay & Dimri, 2025; Fuller-Rowell, 2011). The Earth's ionosphere and thermosphere (IT) form a dynamically coupled system, where physical and chemical processes across various scales interact within the region. Solar wind drivers and the resultant geomagnetic activity induce responses both globally and locally within the IT system (Richmond & Lu, 2000; A. J. Ridley et al., 2006; Fuller-Rowell et al., 1994; Qian et al., 2014; Connor et al., 2016; Mannucci et al., 2016). The response of the IT to geomagnetic activity is both rapid and spatially heterogeneous, often leading to the formation of structures such as storm-enhanced density (SED) plumes and post-sunset depletions (J. C. Foster, 1993; Kelley et al., 2004; J. C. Foster & Vo, 2002a). These phenomena could subsequently result in localized plasma irregularities that influence radio wave propagation (Astafyeva et al., 2014; Hocke & Schlegel, 1996a; Kintner et al., 2007).

Among the most evident midlatitude storm-time responses are SED plumes, which appear as narrow bands of elevated F-region plasma density extending from midlatitudes toward the dayside cusp region (J. C. Foster, 1993; Kelley et al., 2004; J. C. Foster & Vo, 2002a; J. C. Foster & Burke, 2002a; Buonsanto, 1999). SEDs are primarily produced by enhanced electrodynamic transport: penetration and disturbance electric fields drive rapid $\mathbf{E} \times \mathbf{B}$ convection that uplifts and channels midlatitude plasma poleward, often forming a tongue of ionization connecting the dayside ionosphere and polar cap (Deng & Ridley, 2006). In contrast, the depletions that develop equatorward of, or following, SEDs reflect a loss of plasma due to a combination of composition changes, reduced plasma supply from the plasmasphere, and altered transport (Bjoland et al., 2021; Lubyk et al., 2022; Wang et al., 2015; Yang et al., 2015; Zhai et al., 2023). These depletions can extend significantly in latitude and are frequently accompanied by neutral atmospheric disturbances such as traveling atmospheric disturbances (Chang et al., 2022; England et al., 2021; Hocke et al., 1996a, 1996b, 1996c; Hunsucker, 1982a; Otsuka, Suzuki, et al., 2013; Zhang et al., 2022), while background wind dynamics modulate both the SED plume and the surrounding trough structure (Fuller-Rowell, 2011; Richmond & Lu, 2000). These structures are frequently accompanied by atmospheric/ionospheric disturbances, such as TIDs, and are further modulated by background neutral winds and conductivities (Hocke & Schlegel, 1996a; Hunsucker, 1982a; Otsuka, Kotake, et al., 2013; Zhang et al., 2017). These multiscale coupled processes, originating from high-latitude forcing and propagating into lower latitudes, directly impact radio communications, HF/VHF propagation, and Global Navigation Satellite System (GNSS) based navigation and surveillance systems by introducing strong gradients, scintillation, and signal degradation during geomagnetic activity (Kintner et al., 2007; Astafyeva et al., 2014; J. C. Foster, 1993; Coster et al., 2007; Kintner et al., 2009; Priyadarshi, 2015; Lu et al., 2020; Zhang et al., 2019). Understanding these processes requires physically consistent models constrained by comprehensive observations, as they are influenced by dynamic interactions between energetic inputs, neutral winds, plasma redistribution, and composition changes (A. J. Ridley et al., 2006; Meng et al., 2020; Aa, Erickson, et al., 2024; Qian et al., 2014; Schunk et al., 2004a, 2004b; J. C. Foster & Coster, 2007).

TIDs are wave-like electron density perturbations that propagate through the ionosphere with characteristic horizontal wavelengths of a few hundred to a few thousand kilometers and periods from tens of minutes to several hours (Abdu et al., 2003; Hocke & Schlegel, 1996a; Hunsucker, 1982b; Hunsucker & Hargreaves, 2003; Mendillo, 2006; Zhang et al., 2017). They are often driven by atmospheric gravity waves launched at high latitudes or by rapid changes in ionospheric forcing, and they can be ducted or refracted by background winds and composition structures. At midlatitudes, TIDs ride on top of background density enhancements and depletions and can locally sharpen existing gradients, leading to measurable phase and amplitude fluctuations in HF and GNSS signals (Kelley, 2009; Kintner et al., 2009; Priyadarshi, 2015; Skone et al., 2001). Capturing their timing, amplitude, and vertical extent requires both realistic background forcing fields and adequate representation of wave propagation and damping.

The thermospheric composition, particularly the $O/N_2$, plays an essential role in modulating ionospheric electron densities during geomagnetic storms. A reduction in the $O/N_2$ signifies a depletion of atomic oxygen relative to molecular nitrogen, leading to decreased production and/or increased loss of electron density in the ionosphere (Prölss, 1987; Meier, 2021; Luan et al., 2017). Recent NASA GOLD mission observations have provided



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important new constraints on storm-time thermospheric composition changes, including the development, morphology, and temporal evolution of $O/N_2$ depletions during minor and moderate geomagnetic storms (T. Yu et al., 2022; Cai et al., 2024). For example, T. Yu et al. (2022) investigated $O/N_2$ depletion with a latitudinally tilted equatorward boundary during the 20 April 2020 geomagnetic storm, while Cai et al. (2024) examined different $O/N_2$ variation patterns observed by GOLD during the 2–4 August 2021 minor geomagnetic storm. Storm-induced thermospheric expansion causes a redistribution of neutral species, often resulting in significant changes in the $O/N_2$ that directly influence ionospheric electron density variability. Variations in local thermospheric composition alter the balance between ion production and chemical loss processes, thereby contributing to storm-time ionospheric density depletions. Previous studies have shown that storm-time variations in the thermospheric $O/N_2$ strongly influence electron density near the $F_2$ peak, with $O/N_2$ depletions producing enhanced molecular recombination and corresponding reductions in plasma density. These studies demonstrate the important role of thermospheric composition in regulating ionospheric responses during geomagnetic disturbances (Burns et al., 1995; Fuller-Rowell et al., 1991; Mendillo, 2006).

Meanwhile, meridional and zonal neutral winds, together with plasma drifts, influence the spatial and temporal structuring of the storm-time ionosphere (Abdu, 2016; Richmond & Thayer, 2000). The coupling between neutral winds and plasma drifts is mediated through ion-neutral collisions, which transfer momentum and induce plasma convection patterns that shape the ionosphere during disturbed conditions (Richmond & Thayer, 2000). At midlatitudes, enhanced neutral winds driven by high-latitude energy deposition redistribute plasma along magnetic field lines and contribute to the development and evolution of large-scale storm-time features, including storm enhanced density and the midlatitude trough (J. C. Foster & Burke, 2002a; Basu et al., 2008; Heine & Semiannual, 2017; Cherniak et al., 2022).

Over the past decades, numerous efforts have used both observational tools and numerical models to investigate storm-time ionospheric variability (Hocke & Schlegel, 1996b; Schunk et al., 2004a, 2004b; Bust et al., 2004; Aa, Zhang, et al., 2022, 2024; Buonsanto, 1999; Mendillo, 2006; Fuller-Rowell et al., 1991; J. C. Foster & Burke, 2002a; Basu et al., 2008; Astafyeva et al., 2015; Meier, 2021; Lühr et al., 2015; Tsugawa et al., 2007). Atmospheric models such as Thermosphere-Ionosphere-Electrodynamics General Circulation Model (TIE-GCM), The GITM, and Coupled Thermosphere Ionosphere Plasmasphere Electrodynamics model have provided insights into the large-scale responses driven by energetic inputs and electrodynamics (Maute, 2017; A. J. Ridley et al., 2006; Codrescu et al., 2012). In a complementary manner, data assimilation systems such as TEC-Based Ionospheric Data Assimilation System (TIDAS), Global Assimilation of Ionospheric Measurements, Assimilative Mapping of Ionospheric Electrodynamics, and Ionospheric Data Assimilation Three-Dimensional (IDA3D) have integrated sparse observational data to produce reconstructions and nowcasts of ionospheric conditions (Aa, Zhang, et al., 2022, 2024; Bust et al., 2004; Richmond, 1992; Schunk et al., 2004a, 2004b, 2004b). Despite these advances, existing models and assimilation frameworks still face limitations in capturing the full spectrum of storm-time dynamics, particularly small-scale and transient features like TIDs and localized plasma depletions. These features are often highly localized, wave-like, and evolve rapidly, making them difficult to resolve with coarser model grids or assimilation algorithms that tend to smooth or filter transient variability (Decker & McNamara, 2007; Hocke & Schlegel, 1996b). As a result, there is a recognized need for combined modeling and observational approaches that can better reproduce and interpret small-scale storm-time phenomena, leading to a more complete understanding of the ionospheric response. Although significant progress has been made in understanding storm-time ionosphere-thermosphere coupling, accurately representing the relative roles of thermospheric composition, neutral wind dynamics, and electrodynamic forcing remains a continuing challenge for both first-principles and assimilative models. While numerous observational and modeling studies have demonstrated that storm-time $O/N_2$ variations and neutral winds together regulate ionospheric density changes (Buonsanto, 1999; Fuller-Rowell et al., 1991; Mendillo, 2006; Richmond & Thayer, 2000), recent investigations have continued to refine our understanding of these coupled processes using improved observations and modeling capabilities (Kuai et al., 2025; Zhai et al., 2024).

Motivated by these limitations, this study uses the 17 January 2013 geomagnetic storm as a test case to compare a physics-based model with an observation-constrained data assimilation system. Rather than focusing on the uniqueness of this particular storm, our primary objective is to evaluate how well these complementary frameworks reproduce the observed storm-time evolution of the midlatitude ionosphere and thermosphere. By comparing first-principles simulations, assimilative reconstructions, and observations from the Millstone Hill (MH) ISR and GNSS TEC measurements, we identify the similarities and differences between simulation and

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reconstruction in representing large-scale storm morphology, thermospheric composition changes, TIDs, and localized density depletions. This comparison provides insight into the strengths and limitations of physics-based and assimilative approaches for representing storm-time ionosphere-thermosphere coupling and highlights areas where future model and assimilation improvements are needed.

## 2. Methods

### 2.1. Event Window and Analysis Domain

This study focuses on the weak geomagnetic storm of 17–18 January 2013. Geomagnetic activity and upstream solar wind conditions are summarized in Figure 1. This event reached minimum Dst values slightly below −50 nT, Kp values of approximately 4, and auroral electrojet (AE) approaching 900 nT, making it representative of an often occurring weak geomagnetic storm that produces well-defined midlatitude ionospheric responses without the saturation effects often observed during extreme events.

The analyses use a fixed midlatitude domain from 20°–60°N and 230°–300°E to enable consistent regional comparisons where extensive observations are available, focusing on both data-driven reconstructions and physics-based simulations. In addition to full window statistics, targeted snapshots capture key transitions at approximately 16:00, 19:00, 20:10, 22:00, and 23:55 UT on 17 January. Two narrow longitude swaths are used for keogram diagnostics, constructed from GNSS total electron content measurements, to detect TIDs: 265°–270°E and 285°–290°E, sampling the western United States and the sector sampled by the MH ISR. For process-level evaluations, two representative locations are emphasized: a depletion center near 254 °E, 39 °N and the MH site near 288°E, 44°N.

### 2.2. Model, Data Assimilation, and Observational Data

Event simulations were performed using the GITM, a three-dimensional nonhydrostatic first-principles model that self-consistently solves the coupled continuity, momentum, and energy equations governing the ionosphere-thermosphere system (A. J. Ridley et al., 2006; A. Ridley et al., 2026). Unlike empirical or hydrostatic models, GITM explicitly resolves thermospheric dynamics by incorporating key physical forcing processes including solar EUV heating, auroral particle precipitation, ion drag, pressure gradient forcing, viscosity, neutral drag, thermal conduction, gravity, and Coriolis effects. In this study, solar irradiance forcing is specified using the Flare Irradiance Spectral Model (FISM v1.1), while the high-latitude electrodynamic forcing is specified using the Weimer (2005) electric potential model and auroral precipitation from the Feature Tracking of Aurora (FTA) model (Wu et al., 2021; Wu & Ridley, 2024). The physical parameters and forcing configurations used in this study are summarized in Table 1 of Nwankwo and Ridley (2025). Simulations use a horizontal grid resolution of 4° longitude by 2° latitude with 50 vertical altitude levels spanning 100–600 km. For analysis, global model outputs are vertically integrated to total electron content and subset to the 20° to 60° N and 230° to 300° E domain.

To complement first-principles model simulations, we use reconstructions from the TEC-Based Ionospheric Data Assimilation System (TIDAS), a regional three-dimensional ionospheric assimilation framework developed to reconstruct time-evolving electron density structures over North America (Aa, Zhang, et al., 2022). TIDAS assimilates multiple heterogeneous observational data sources, including dense ground-based GNSS line-of-sight total electron content measurements, COSMIC radio occultation observations, JASON satellite altimeter TEC data, and electron density profiles measured by the MH ISR. The system employs a hybrid Ensemble-Variational assimilation framework, in which background error covariance structures are estimated from an ensemble of ionospheric states generated from corrected NeQuick model outputs, while a three-dimensional variational (3DVAR) assimilation scheme optimally combines observational constraints with model background estimates. The resulting reconstructions provide electron density distributions with a spatial resolution of 1° × 1° horizontally, 20 km vertical resolution, and a temporal cadence of 5 min, enabling the resolution of storm-time large-scale ionospheric structures such as storm-enhanced density plumes and associated regional electron density gradients.

Because MH ISR measurements are assimilated as one of several observational constraints within TIDAS, the assimilation product is not fully independent of the ISR observations used later for comparison. However, the ISR profiles are incorporated together with a substantially larger GNSS TEC observational network over a relatively coarse assimilation grid, preventing one-to-one reproduction of individual ISR structures. In practice, TIDAS

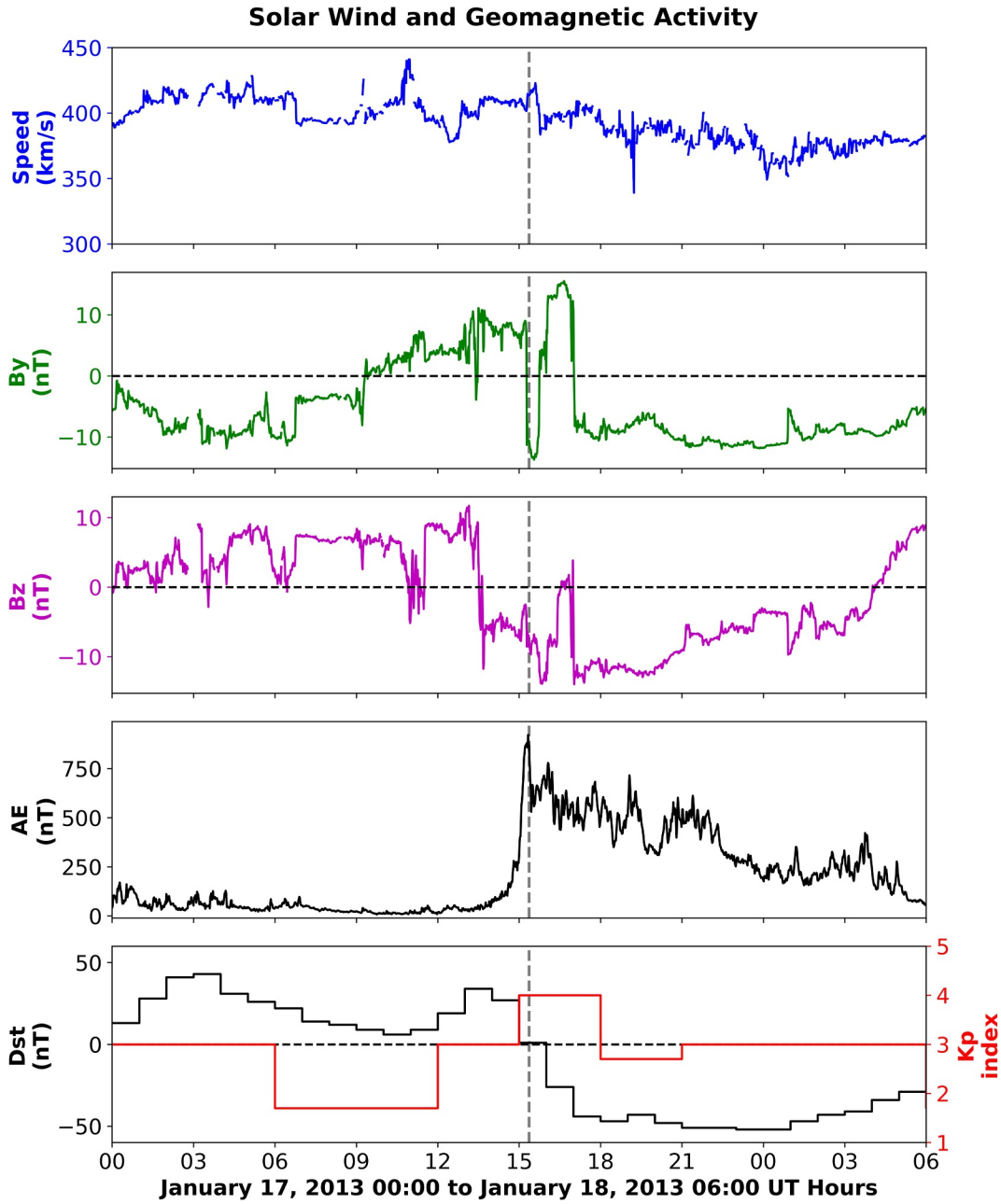



**Figure 1.** Solar wind and geomagnetic activity during 17–18 January 2013. From top to bottom: (1) solar wind speed, (2) interplanetary magnetic field (IMF) $B_y$, (3) IMF $B_z$, (4) auroral electrojet index, and (5) Dst index (black, left axis) with the Kp index (red, right axis). The vertical dashed line marks the approximate onset of the main storm phase near 15 UT on 17 January 2013.

smooths short-lived small-scale variability while preserving the dominant large-scale storm morphology and background ionospheric structure. This characteristic makes TIDAS particularly useful for evaluating large-scale storm-time responses and mean state evolution, while also allowing assessment of how much mesoscale TID activity and depletion structures present in both GITM simulations and ISR observations.

To compute the $O/N_2$ in GITM, the following technique was used. Let $N_2(z)$ denote the simulated $N_2$ number density as a function of altitude ($z$). For each grid point, we define a fixed reference $N_2^i$ column integral as:



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$$N_2^i = 10^{21} = \int_{Alt}^{\infty} N_2(z)\,\partial z \tag{1}$$

and solve for the lower integration altitude $Alt$. Note that the integrated quantity $N_2^i$ is held constant, so $Alt$ necessarily increases as the atmosphere expands and the same $N_2^i$ column is reached at a higher geometric altitude. The middle panel in Figure 6 shows the resulting $Alt$, and can be used as a proxy for storm time thermospheric expansion.

The corresponding $O^i$ column integral is then computed:

$$O^i = \int_{Alt}^{\infty} O(z)\,\partial z \tag{2}$$

and an integrated composition ratio is computed

$$\left(\frac{O}{N_2}\right)^i = \frac{O^i}{N_2^i}. \tag{3}$$

Since $N_2^i$ is again constant by construction, variations in $(O/N_2)^i$, referred to as $O/N_2$ from now on, primarily reflect changes in the O abundance relative to $N_2$ at a constant $N_2$ column integral.

## 3. Results and Discussion

To place the ionospheric response in context, Figure 1 shows the solar wind conditions and geomagnetic activity during 17–18 January 2013. Solar wind speed remained relatively steady between approximately 390 and 440 km/s before gradually decreasing to below 380 km/s later in the event. The interplanetary magnetic field (IMF) components exhibited substantial variability, with $B_y$ transitioning from predominantly negative to positive values before returning negative, while $B_z$ turned southward near 15 UT and remained predominantly negative for several hours. The sustained southward IMF $B_z$ provided favorable conditions for enhanced solar wind-magnetosphere coupling and marked the onset of the main storm phase analyzed in this study.

The AE index increased rapidly to nearly 900 nT shortly after 15 UT, indicating intensified AE activity and enhanced high latitude energy input into the ionosphere-thermosphere system. At the same time, the Dst index transitioned from a weakly positive excursion to a sustained decrease, reaching values slightly below −50 nT during the main phase before gradually recovering. The Kp index increased from quiet conditions to approximately 4 during the disturbed interval, consistent with a weak geomagnetic storm. Throughout the event, the daily F10.7 solar flux remained nearly constant at 120 sfu, indicating relatively stable solar EUV conditions. Together, these observations indicate that the 17 January 2013 event was a weak geomagnetic storm driven by southward IMF conditions.

### 3.1. Comparison of GITM and TIDAS

We first compare the storm-time evolution of TEC and ionospheric NmF2 from GITM and TIDAS on 17 January 2013. Figure 2 shows snapshots at 10, 13, 16, 19, and 22 UT. At 10 UT, GITM shows that the nightside has very low density, while TIDAS shows an unstructured density. Both show the edge of the auroral zone at the highest latitudes. By 13 UT, in correspondence with the dawn sector, GITM shows consistency with TIDAS in both magnitude and spatial distribution of TEC, with both showing a gradient from the dayside in the east to the nightside in the west, although TIDAS shows a larger TEC on the nightside. During the storm main phase at 16 UT, enhanced densities appear in both, with a westward shift in GITM relative to TIDAS. At 19 UT, GITM agrees well in both magnitude and width parameters, with both having enhanced densities in the west, a prominent trough in the northeast corner, and a storm enhanced density feature equatorward of the trough and the north-central region. By 22 UT, both GITM and TIDAS show consistent features, including the enhanced trough and a general decrease in density in the east relative to earlier times. Although the structures are narrower and contain more fine-scale variability, the model and assimilation agree on the main spatial patterns.



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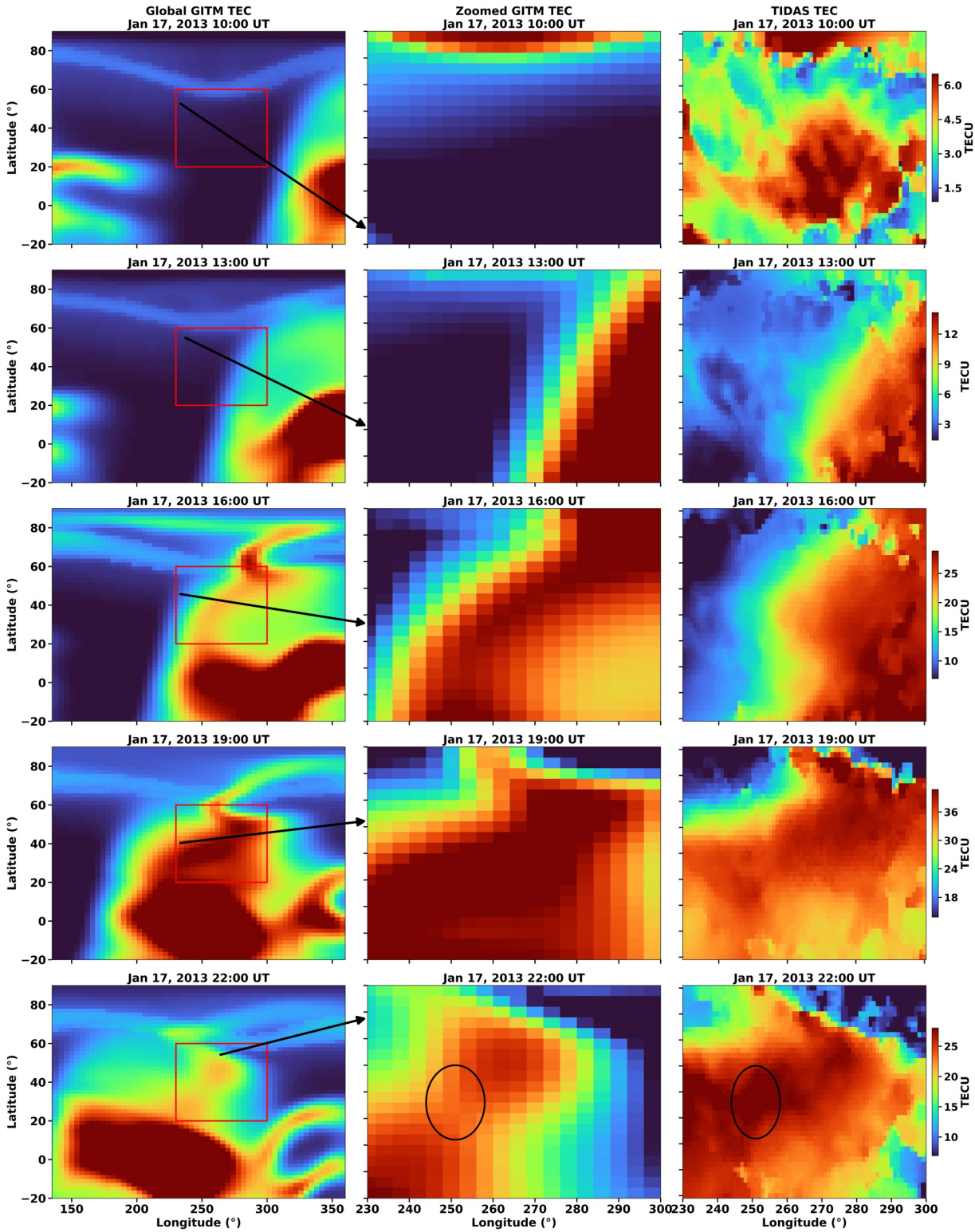



**Figure 2.** Comparison of TEC from Global Ionosphere Thermosphere Model (GITM) and TIDAS on 17 January 2013 at (from top to bottom) 10, 13, 16, 19, and 22 UT. The left column shows GITM TEC over the longitude range 135°–360°E and latitude range 20°S–90°N. The red box marks the TIDAS reconstruction domain (20°–60°N, 230°–300°E). The middle column shows the corresponding GITM subset within this domain, and the right column shows the corresponding TIDAS TEC reconstruction. The black ellipses in the 22:00 UT panels identify a localized region where TIDAS shows enhanced TEC while GITM produces a relative TEC reduction.



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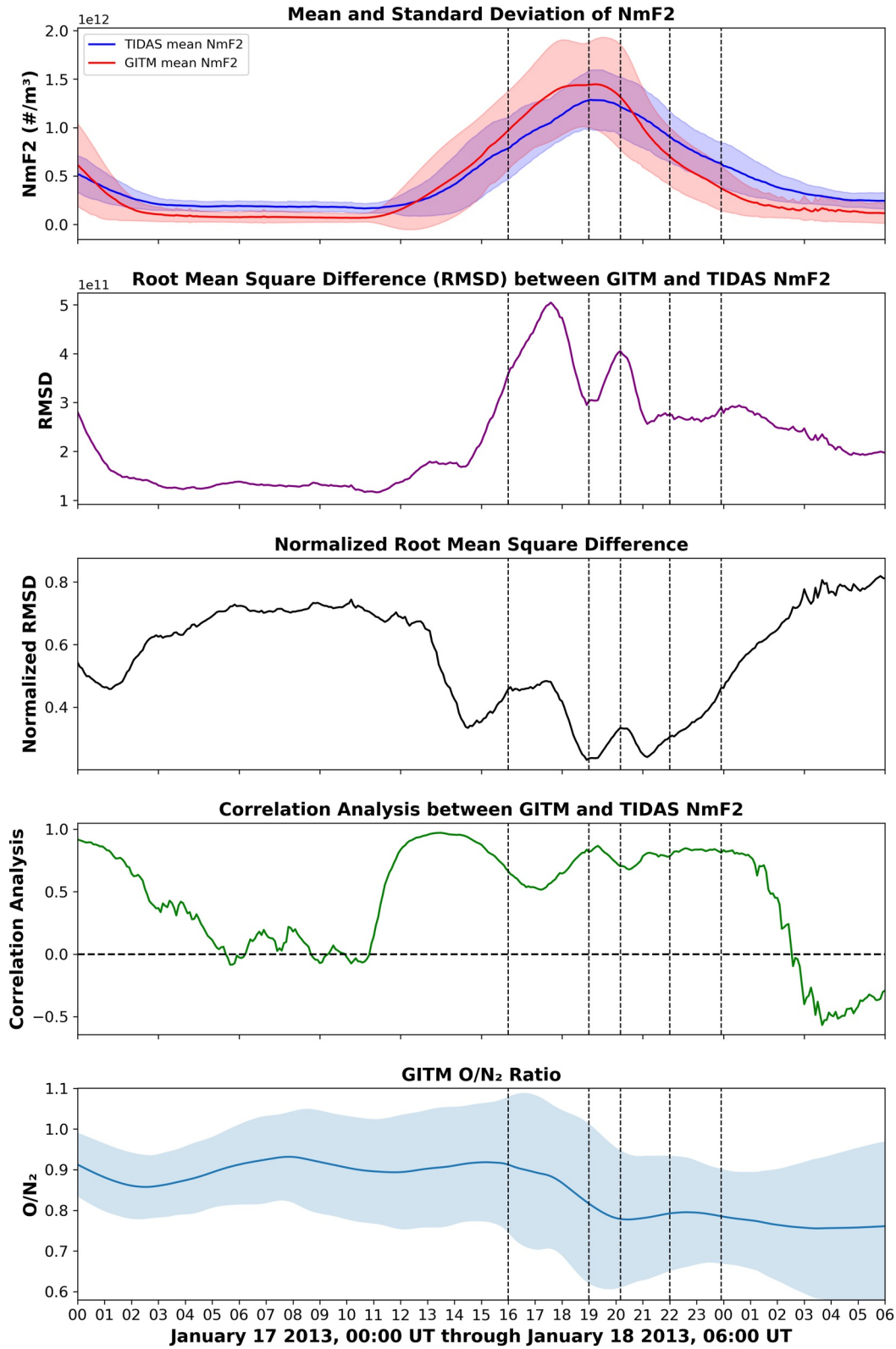



**Figure 3.** Statistical comparisons of NmF2 between Global Ionosphere Thermosphere Model (GITM) and TIDAS over the TIDAS domain (20°–60° N, 230°–300°E) from 00 UT on 17 January to 06 UT on 18 January 2013. Panels show (top to bottom): mean and standard deviation of NmF2, root mean square difference (RMSD), normalized RMSD, correlation analysis, and the GITM O/$N_2$. Vertical dashed lines mark 16, 19, 20:10, 22, and 23:55 UT.



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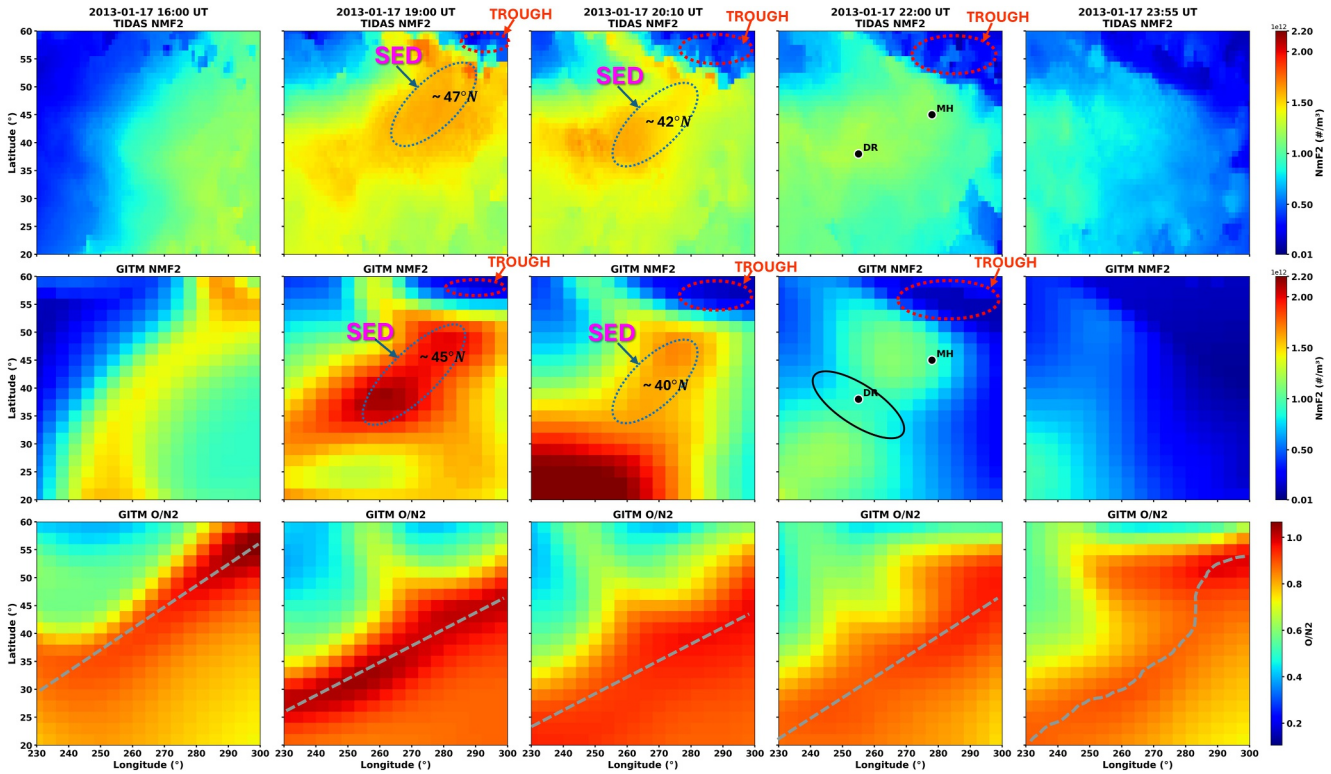



**Figure 4.** Comparison of $NmF_2$ from TIDAS (top row) and Global Ionosphere Thermosphere Model (GITM) (middle row) with the GITM $O/N_2$ (bottom row) on 17 January 2013 at 16:00, 19:00, 20:10, 22:00, and 23:55 UT. The storm enhanced density (SED) and trough are indicated in the $NmF_2$ panels, while the dashed lines in the bottom row indicate the peak of the $O/N_2$ composition front. At 22:00 UT, the depletion region (DR) in GITM is labeled DR, and the location of the Millstone Hill radar measurements is labeled MH.

To quantify these similarities and discrepancies, Figure 3 presents comparisons of the peak electron density (NmF2) between GITM and TIDAS, including the mean and standard deviation of NmF2 spatially averaged over the selected TIDAS analysis region, together with root mean square difference (RMSD), normalized RMSD, correlation analysis, and the model-derived $O/N_2$. Each of these quantities were calculated over the selected TIDAS domain at each 5-min interval.

At night, before 12 UT, GITM underestimates NmF2 compared to TIDAS, resulting in an elevated normalized RMSD. The lack of a plasmaspheric downwelling of the F-region at mid-latitudes within GITM is the driving factor in this underestimation. The correlation value is also low on the nightside due to a very low density in GITM, providing almost no structure. Hence, the nightside midlatitude ionosphere is not accurately represented in GITM. As dawn approaches, the GITM-TIDAS correlation rises above 0.5 and remains elevated through much of the day. In addition, the normalized RMSD goes down, mostly driven by the increased density. The RMSD has multiple peaks around 18 UT, which coincides with the onset of the storm main phase. These peak differences could be due to shifts in features between the two. For example, at 22:00 UT in Figure 2, the GITM and TIDAS results show broadly similar large-scale TEC structures. However, within the regions highlighted by the black ellipses, TIDAS shows a localized TEC enhancement near 255°E and 30°N (approximately 250°–260°E and 20°–40°N), whereas GITM simulates a TEC reduction over the same region. This localized spatial offset/mismatch contributes to an increased RMSD. Later in the day, the NmF2 from GITM switches from being systematically higher than TIDAS to being lower than in TIDAS, consistent with the concurrent decline in the GITM $O/N_2$. The decrease in NmF2 after about 20 UT may reflect the combined effects of local time progression and composition-driven changes.

To further compare the spatial evolution of the storm response for the electron density over the United States sector, Figure 4 presents $NmF_2$ from TIDAS and GITM together with the corresponding GITM $O/N_2$. At 16:00 UT, both TIDAS and GITM show the developing storm-time enhancement at midlatitudes. By 19:00 and 20:10



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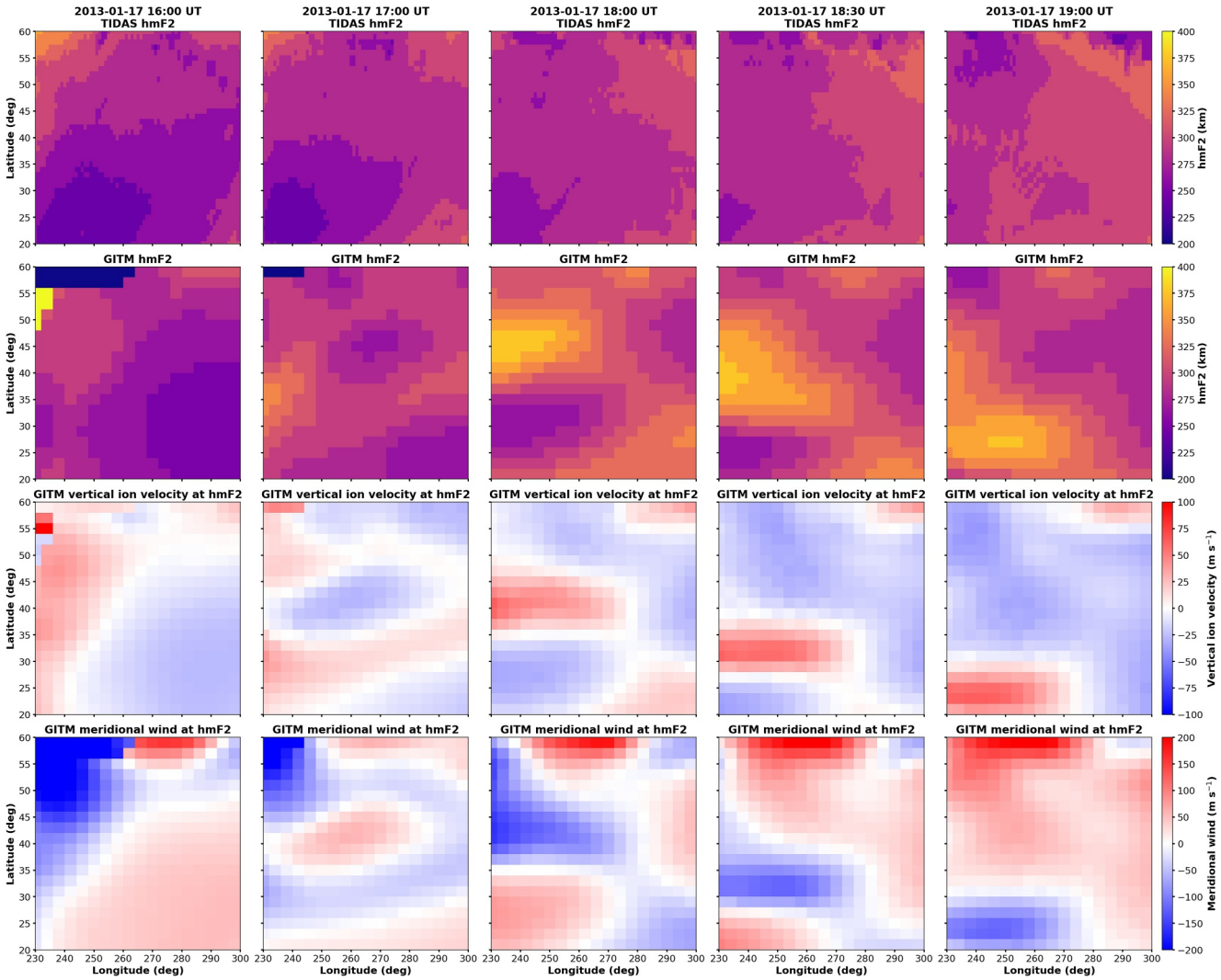



**Figure 5.** Latitude–longitude maps of $hmF_2$ from TIDAS (first row) and Global Ionosphere Thermosphere Model (GITM) (second row), GITM vertical ion velocity evaluated at $hmF_2$ (third row), and GITM meridional neutral wind evaluated at $hmF_2$ (fourth row) at 16:00, 17:00, 18:00, 18:30, and 19:00 UT on 17 January 2013. Positive vertical ion velocity represents upward plasma motion, while negative meridional wind represents equatorward (southward) flow. From 16:00 to 19:00 UT, the regions of equatorward meridional wind and upward ion motion remain spatially aligned as they progress equatorward. The same progression is evident in the region of elevated GITM $hmF_2$, indicating that meridional-wind-driven uplift contributes to the formation and equatorward evolution of the SED plume seen in Figure 4.

UT, both reproduce a distinct SED-related $NmF_2$ enhancement, although the exact location and fine-scale structure differ. The enhanced feature shifts from approximately 47° to 42°N in TIDAS and from approximately 45° to 40°N in GITM.

By 22:00 UT, the large-scale storm response transitions from enhancement to depletion. Both TIDAS and GITM show a marked reduction in $NmF_2$, although GITM develops a more structured, wave-like depletion pattern than TIDAS. At the same time, the modeled $O/N_2$ composition front continues its equatorward progression (Figure 4), while the regional mean $O/N_2$ decreases after approximately 18 UT, as shown in Figure 3. The concurrent decreases in $O/N_2$ and $NmF_2$ indicate that thermospheric composition becomes increasingly important during this later phase of the storm response. The reduced $O/N_2$ favors increased molecular recombination and is therefore consistent with the broad decrease in electron density across the study region. By 23:55 UT, the electron density is substantially reduced in both systems, with a trough remaining in the northeast region. The physical processes associated with the initial SED enhancement and the subsequent broad depletion are examined separately in the following subsections.



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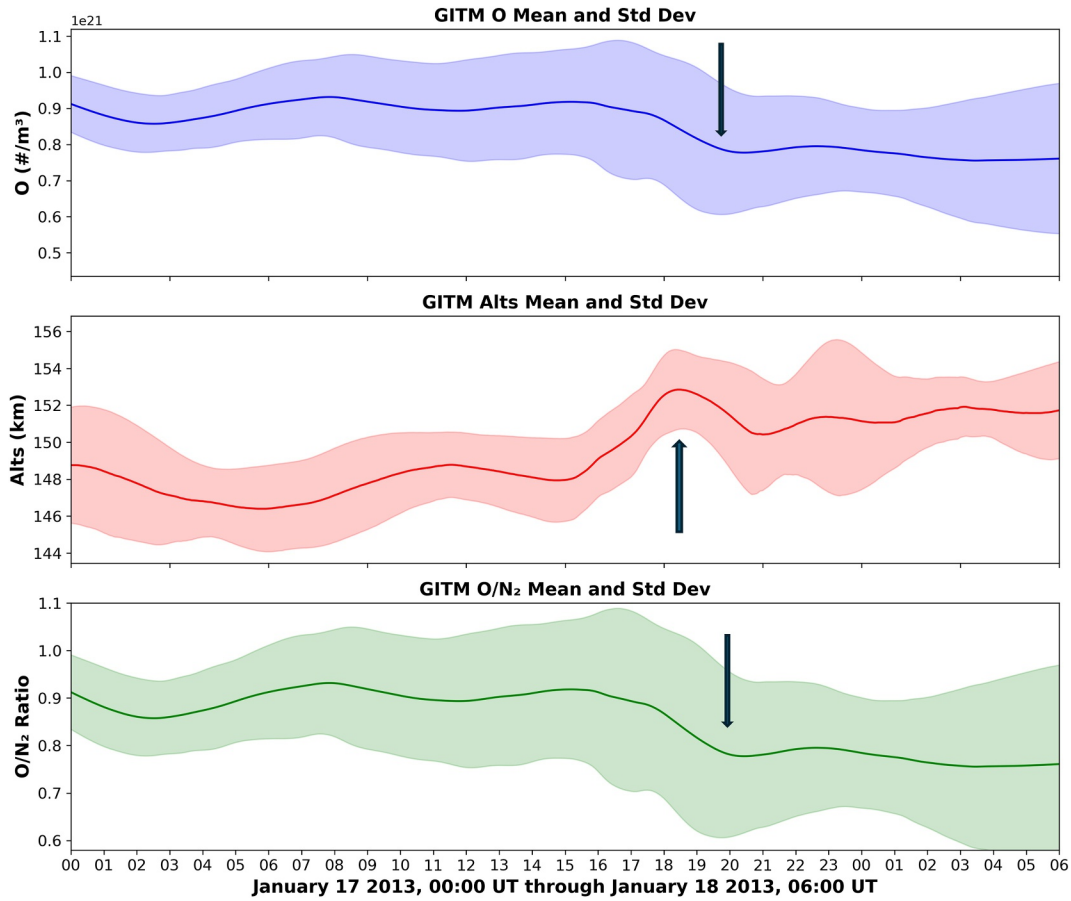



**Figure 6.** Temporal evolution of GITM-simulated (top) atomic oxygen (O) density, (middle) effective altitude of the $N_2$ column integral, and (bottom) $O/N_2$ over the TIDAS field of view (20°–60° N, 230°–300°E) from 00 UT on 17 January to 06 UT on 18 January 2013.

### 3.2. SED Formation

The initial phase of the storm response is characterized by the development and equatorward progression of the storm enhancement density (SED) between approximately 16:00 and 19:00 UT. Figures 2 and 4 show that the enhancement develops in both GITM and TIDAS, indicating that it is a large-scale feature of the storm-time ionosphere rather than a model-specific structure. To examine the mechanism associated with its formation, Figure 5 compares $hmF_2$ from TIDAS and GITM with the GITM vertical ion velocity and meridional neutral wind at 16:00, 17:00, 18:00, 18:30, and 19:00 UT.

From 16:00 to 19:00 UT, GITM shows an increase in $hmF_2$ that is consistently associated with upward ion motion, most likely driven by equatorward meridional winds. At 16:00 UT, equatorward winds occur primarily between approximately 35° and 60°N and 230° and 260°E. Upward ion velocities occur over approximately the same region, which also corresponds to elevated $hmF_2$. Outside this region, the meridional winds are predominantly poleward, and the vertical ion velocities are generally downward. The spatial correspondence among the meridional wind, vertical ion velocity, and $hmF_2$ persists at 17:00 and 18:00 UT, although the associated structures shift progressively equatorward. By 18:30 UT, the equatorward wind is concentrated between approximately 28° and 38°N and 230° and 270°E. Upward ion motion occupies approximately the same region, while poleward winds and downward ion motion dominate much of the surrounding domain. The region of elevated GITM $hmF_2$ also shifts into this latitude and longitude range.

By 19:00 UT, the equatorward wind and upward ion velocity have moved farther equatorward and remain colocated with the region of elevated $hmF_2$. This repeated spatial correspondence is physically consistent with equatorward neutral winds driving plasma upward along inclined magnetic field lines in the Northern

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Hemisphere, thereby raising the F2 layer. The progression of this wind-driven uplift also follows the equatorward development of the SED-related enhancement identified in Figure 4. This indicates that the SED plume is most likely driven by meridional winds that elevate $hmF_2$ within the region. Raising the plasma to higher altitudes, where molecular recombination is slower, allows the enhanced plasma density to be maintained as the feature progresses equatorward. Storm-time electrodynamic transport and horizontal plasma redistribution may also contribute to the broader SED morphology, but the spatial and temporal correspondence in Figure 5 identifies meridional-wind-driven uplift as the principal mechanism associated with the initial enhancement examined here.

After approximately 20 UT, the storm response transitions from this wind-driven enhancement phase to a broad reduction in electron density. In addition, there is a weak downward ion velocity during this time. This would further drive a reduction in the $NmF_2$, since it would push the ions toward the molecular rich neutrals at lower altitudes. The combined role of this downward transport and thermospheric composition in producing the later depletion is examined in the following subsection.

### 3.3. Changes Driven by Thermospheric Composition

Following the wind-driven formation and equatorward progression of the SED plume, the large-scale ionospheric response transitions to a broad decrease in electron density. By 22:00 UT, both TIDAS and GITM show a marked reduction in $NmF_2$. This reduction occurs concurrently with the equatorward progression of the modeled $O/N_2$ composition front in Figure 4 and the decrease in regional mean $O/N_2$ shown in Figure 3. The temporal correspondence indicates that thermospheric composition becomes increasingly important during the later phase of the storm response. A reduction in $O/N_2$ increases the relative importance of molecular species and promotes faster ion loss through molecular recombination. The evolving composition therefore provides the clearest explanation for the broad regional decrease in electron density after the SED enhancement.

To investigate how storm-time thermospheric composition influences the evolution of the ionosphere within the study region, we examine the temporal evolution of GITM simulated column integrated atomic oxygen $(O^i)$, the mean simulated altitude of the $N_2$ integral ($Alt$), and a corresponding column $O/N_2$ over the TIDAS analysis domain (Figure 6). Rather than describing the global morphology of the storm, these quantities provide a regional measure of the evolving background thermospheric state that influences the modeled ionospheric response discussed in the previous section.

During the early prestorm quiet period, atomic oxygen density remained relatively steady, with only a modest increase toward the storm onset. However, after 18–19 UT, the $O^i$ density decreased in coincidence with the storm main phase. At the same time, the mean simulated altitude of the $N_2$ integral shifted upward. This upward shift reflects the expansion of the thermosphere under storm forcing. This drove a decrease in integrated $O^i$, leading to a decreased $O/N_2$ that is in agreement with the timing of the reduced NmF2 values seen in GITM relative to TIDAS.

To determine how realistic the simulated GITM composition changes are, Figure 7 compares GUVI satellite-derived $O/N_2$ with GITM $O/N_2$ along the 10.2 LT ascending sector. Both data sets show a daytime mid-latitude perturbation and an associated equatorward progression of reduced $O/N_2$ during the main phase, indicating that GITM captures the overall magnitude and timing of the storm-time composition response. However, the depleted region in GITM extends several degrees farther equatorward than in GUVI, suggesting that the equatorward transport of composition in the model is too strong. Its existence points to a potential effect of meridional neutral winds that are somewhat stronger in amplitude than in reality.

Figure 6 demonstrates that storm-time thermospheric expansion is accompanied by a gradual decrease in the regional column-integrated $O/N_2$ over the TIDAS analysis domain. These composition changes are consistent with the later reduction in NmF2 simulated by GITM and indicate that storm-time thermospheric composition contributes to the temporal evolution of the ionosphere within the study region. Similar composition-driven ionospheric responses have been reported in previous observational and modeling studies (Buonsanto, 1999; Fuller-Rowell et al., 1991; Meier, 2021; Mendillo, 2006; Prölss, 1987).

At smaller scales, both systems capture a pronounced trough adjacent to the SED, most likely associated with fast westward ion drifts in a subauroral polarization stream (SAPS) configuration. This feature has been extensively documented using MH ISR observations and coordinated measurements (J. C. Foster & Burke, 2002b; J. C. Foster

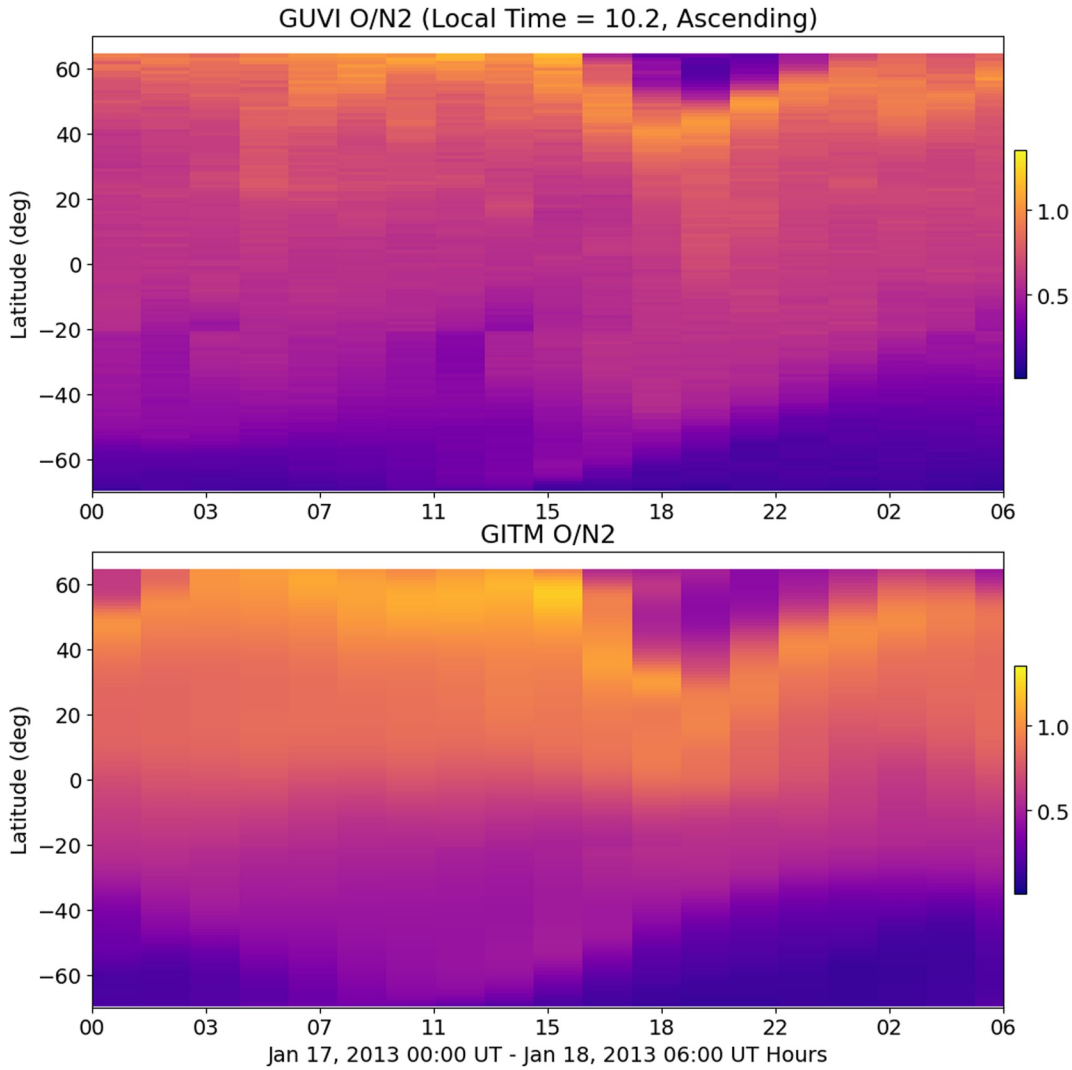



**Figure 7.** Latitude–time comparison of $O/N_2$ from GUVI (top) and GITM (bottom) along the 10.2 LT ascending sector from 00 UT on 17 January to 06 UT on 18 January 2013.

& Vo, 2002b; J. Foster et al., 2007; Aa et al., 2020). More recent multi-instrument and data-assimilative studies have further characterized the occurrence, evolution, and morphology of SAPS and SED over the North American sector (Aa, Dzwill, et al., 2024; Aa et al., 2023). Beyond the SAPS-associated depletion, the GITM $O/N_2$ distribution exhibits additional small-scale wave-like structures superimposed on the large-scale composition pattern. These features suggest that thermospheric neutral wind variability may also contribute to localized ionospheric structuring. Mesoscale TIDs and localized density structures associated with neutral wind variability are superimposed on this large-scale evolution during the 17 January 2013 storm and are examined separately in the following subsection.

### 3.4. Changes Driven by Thermospheric Wind Variations

While Figure 6 characterizes the regional evolution of the background thermospheric composition, it does not explain the localized depletion structures that develop within the domain. We therefore next examine the spatial morphology of NmF2 to investigate the physical processes associated with these localized features. Figure 4 highlights the spatial distribution of NmF2 at 22 UT, when the GITM simulation shows a distinct region of depletion (depletion region (DR)), while TIDAS does not indicate this depletion, showing a much smoother pattern.



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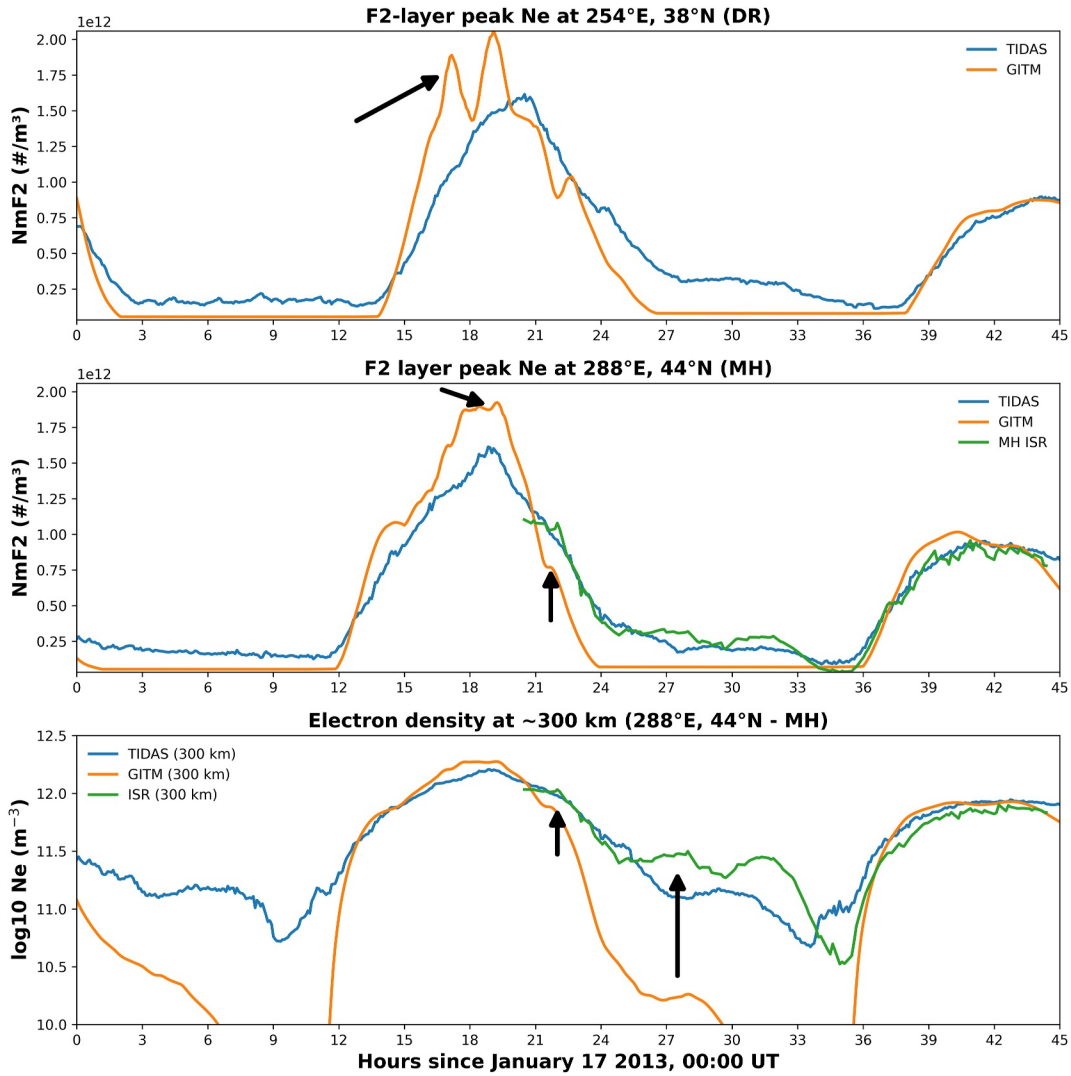



**Figure 8.** (Top) Close-up view of the depletion region in NmF2 at 22 UT on 17 January 2013 from Global Ionosphere Thermosphere Model (GITM) and TIDAS. (Middle) NmF2 comparison at the Millstone Hill (MH) site, located near the edge of the modeled depletion. (Bottom) Time series of electron density at ∼300 km altitude from MH-ISR, GITM, and TIDAS, shown in hours since 00 UT on 17 January 2013 (0–45 hr corresponds to 17–18 January). The major descending wavefronts are identified using guide lines to facilitate visual comparison between the ISR observations and GITM.

This difference raises two questions: (a) what produces the localized depletion in GITM, and (b) why is a corresponding structure not evident in TIDAS? One possible explanation is that localized thermospheric wind variations redistribute plasma along magnetic field lines and generate the small-scale electron-density variability resolved by GITM.

The lack of comparable structure within TIDAS could arise from either physical or methodological causes. One possibility is that the small-scale, wave-like perturbations (e.g., TIDs-like structures) that contribute to the depletion seen in GITM are real but are smoothed out in the assimilation by the spatial and temporal filtering inherent to the TIDAS framework. Alternatively, the absence of the depletion could reflect a lack of such small-scale structures in the data used for assimilation.

In order to understand whether the disagreement observed in Figure 4 at 22 UT is driven by GITM showing inaccurate detailed structuring or TIDAS's inability to capture small-scale structures, we compare the results at the DR and Millstone Hill (MH) mid-latitude locations. To better characterize this feature, Figure 8 provides line plots of the density at these two locations.



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The top panel of Figure 8 shows clear wave-like structures in the GITM-simulated electron density at the DR site. However, there are no simulated or measured large-amplitude wave-like structures (middle panel) at MH. Around 21 UT in the middle panel, there are small hints of waves but no large features, such as those observed in the DR. At the MH site, both GITM and TIDAS exhibit similar temporal trends without evidence of strong depletions, which lends to the conclusion that the discrepancy seen in the top panel of Figure 8 is spatially confined. Comparing results between the DR and MH shows that storm-time ionospheric variability can vary significantly across relatively short distances, with localized structures superimposed on the large-scale background response.

The bottom panel of Figure 8 isolates the time dynamics at ∼300 km altitude in electron density for the ISR, GITM, and TIDAS on a log-scale. The MH ISR trace represents a direct, instrument-based measurement along a narrow beam, and shows fine-scale variability in electron density that is not reproduced in TIDAS. This is expected: although TIDAS assimilates MH ISR data, it does so together with a much larger GNSS TEC data set on a coarser grid and with temporal and spatial smoothing, so the detailed space–time structure seen by the ISR is absorbed into a broader background and not retained one-to-one in the reconstruction. Because these intervals occur near dusk, the nighttime GITM $N_e$ is too low due to rapid recombination and lack of plasmaspheric refilling, so the later post-midnight structure seen by the ISR is not sustained in the model. However, both GITM and the ISR show wave-like structures around 22 and 28 UT. The ISR measures another crest of density around 32 UT, but by that time the GITM nighttime density is too low to show structuring at all, while TIDAS continues to represent only the large-scale background.

To further evaluate the small-scale storm-time variability, Figure 9 shows the temporal evolution of electron density profiles from ISR, GITM, and TIDAS between 20 UT on 17 January and 20 UT on 18 January. Reiterating, while TIDAS assimilates MH ISR data, the radar measurements provide a high-resolution, instrument-based reference along a single line of sight, whereas TIDAS represents a smoothed, regional reconstruction constrained by ISR together with GNSS TEC and other data.

The ISR data show wave-like structures between approximately 20 UT on 17 January and 10 UT on 18 January, characterized by alternating enhancements and depletions extending from about 200 to 500 km altitude. These quasi-periodic features descend in altitude with time, a signature commonly associated with TIDs driven by storm-time forcing (Borries et al., 2009; Kogure et al., 2024; Nicolls et al., 2004). Both MH ISR and GITM show descending wave structures, particularly near dusk and local midnight. Although GITM reproduces the timing and vertical extent of the earlier wave packets, it underestimates the nighttime electron densities. As a result, the fourth wave-like structure observed by the MH ISR around 08 UT on the 18th does not appear in the model, because the simulated nighttime densities are too low to sustain detectable perturbations. This highlights that GITM reproduces the downward phase progression of the first several TIDs, whereas later wave packets are absent because the simulated nighttime electron density is too depleted.

In contrast, TIDAS, by its nature, provides a smoother temporal evolution of Ne with limited evidence of wave activity. The lack of small-scale oscillations in particular suggests that the assimilation procedure filters out small-scale, transient variations that are present in both observations and model results. The comparison, therefore, indicates that the wave structures simulated by GITM and observed by ISR are physically realistic, while TIDAS struggles to capture such dynamic features. These results also provide confidence that GITM is likely capturing the TIDs correctly, at least in the MH area and in the afternoon to post-dusk time frame.

To assess whether the wave activity inferred from the ISR and simulated by GITM also appears in vertically integrated measurements, we compare detrended TEC (dTEC) keograms from GITM calculations and from GNSS data along two longitude sectors (265°E–270°E and 285°E–290°E) during 17–18 January 2013 (Figure 10). This is a common method for exploring wave activity within the ionosphere (Trop et al., 2025; Zhang et al., 2019, 2022). In Figure 10, the GITM dTEC fields are compared with GNSS keograms derived using a 60-min sliding-window high-pass filter, which removes background TEC variations and emphasizes large-scale, low-frequency TID and AGW components. We note that the nearly vertically aligned structures in the GNSS dTEC keograms indicate an approximately simultaneous response across all latitudes. Similar coherent dTEC responses have been interpreted as signatures associated with penetration electric fields (Zhang et al., 2023). Therefore, these structures are consistent with a possible prompt penetration electric field contribution, although they are not uniquely diagnostic of PPEF in the absence of direct electric field or vertical plasma drift measurements. The Weimer (2005) electrodynamic forcing used in GITM does not explicitly represent low and midlatitude PPEF effects, which explains the absence of comparable vertically aligned perturbations in the

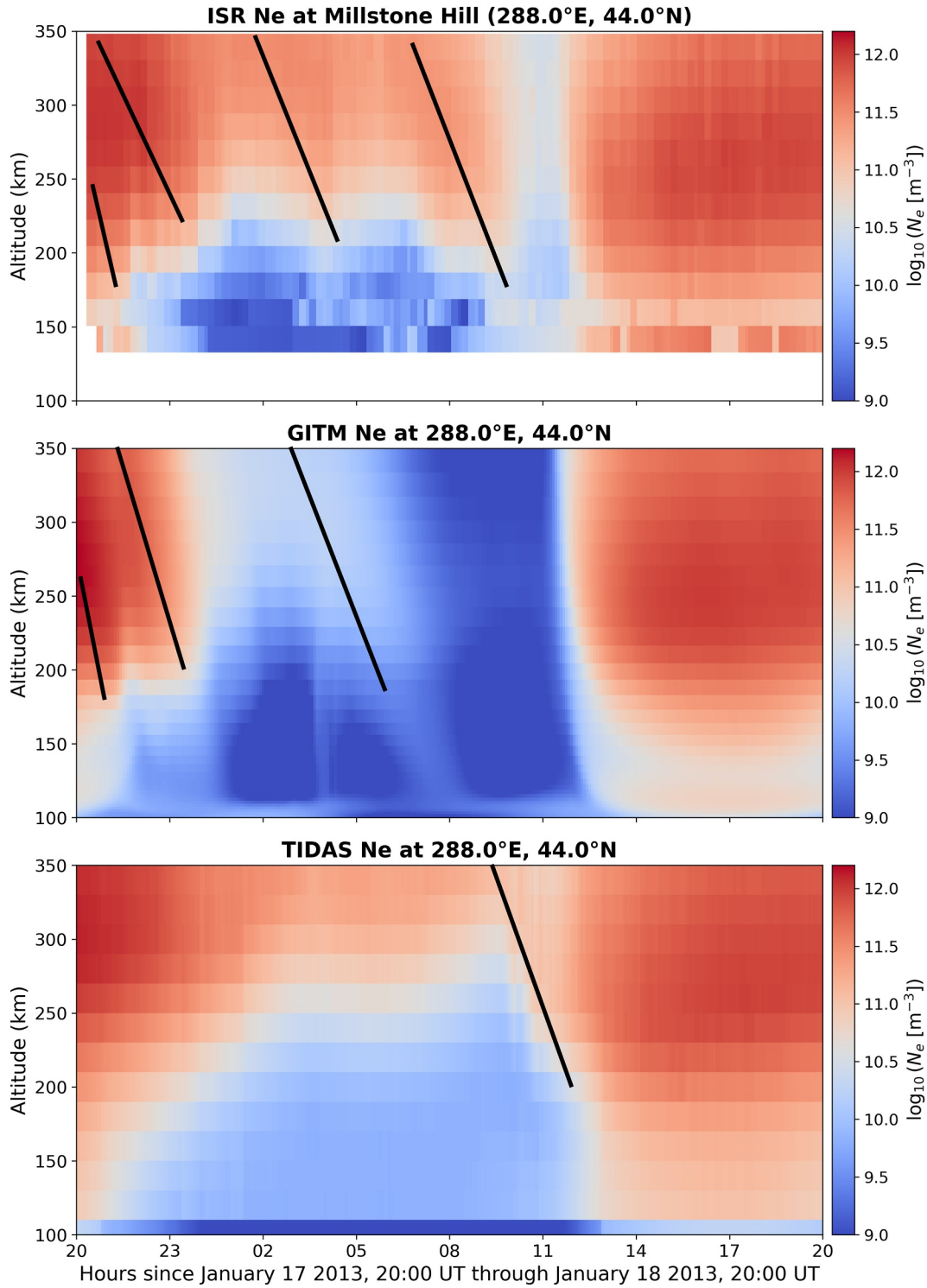



**Figure 9.** Temporal evolution of electron density profiles from the Millstone Hill incoherent scatter radar, GITM, and TIDAS between 20 UT on 17 January and 20 UT on 18 January 2013.

simulation. Both data sets show discernible large-scale TIDs after the main storm phase (around 21 UT), but GNSS dTEC amplitudes reach only about 0.5 TECU, significantly smaller than the GITM predictions. Radar observations by Zhang et al. (2017) reported roughly 20% TID amplitudes during more intense storms, consistent with the classification of this event as a weak geomagnetic storm.



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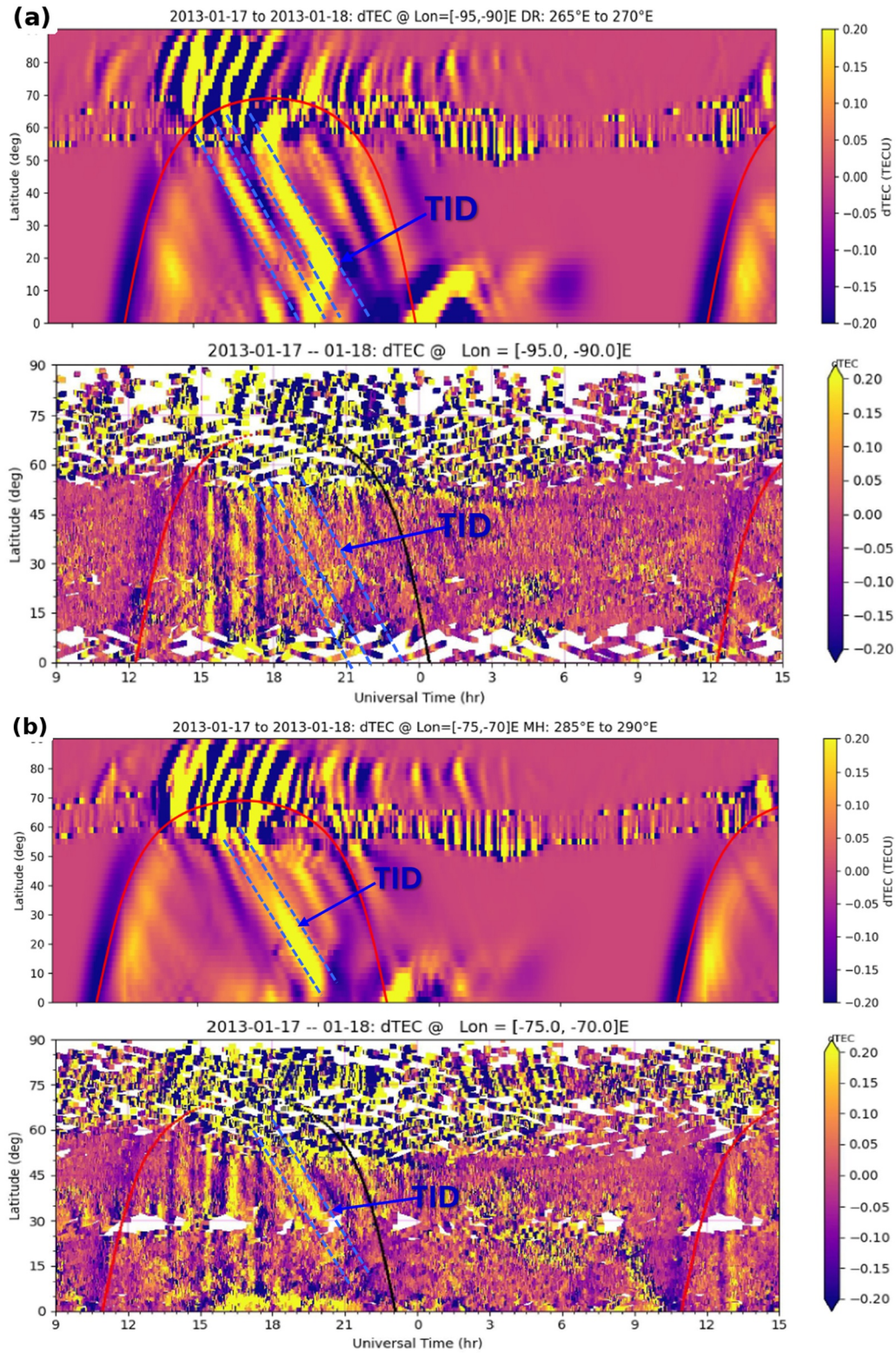



**Figure 10.** Comparison of detrended TEC (dTEC) variations from Global Ionosphere Thermosphere Model (GITM) simulations and GNSS observations during the 17–18 January 2013 geomagnetic storm. Panel (a) shows the GITM dTEC keogram for the 265°E–270°E sector above the corresponding GNSS dTEC keogram. Panel (b) presents the same comparison for the 285°E–290°E sector. The blue dashed lines identify the equatorward-moving TID structures, while the red curves indicate the solar terminator.



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In both sectors, banded structures in dTEC reveal large-scale, low-frequency TIDs after the main phase. The 285°E sector (i.e., MH sector) shows relatively weaker perturbations in GITM than the 265°E sector, where GITM exhibits stronger and more persistent wave activity than GNSS. This discrepancy indicates a longitude-dependent bias in GITM. Amplitudes in GNSS (≤∼0.5 TECU) are smaller than in GITM and consistent with the smoother behavior seen in TIDAS. This suggests that the model may over-amplify disturbances at western longitudes during this storm.

The structure simulated by GITM near 22 UT likely corresponds to TIDs originating from higher latitudes. The contrasting longitudinal responses imply that the coupling between meridional neutral winds, ion drifts, and thermospheric composition (which differs between longitude sectors) controls the strength and spatial variability of wave propagation. To investigate the physical drivers of the wave activity observed in GITM and the ISR, Figures 11 and 12 show GITM-simulated electron density, meridional neutral winds, vertical ion drifts, and vertical neutral winds at two fixed geographic locations: within the DR and at the MH site, respectively. The analysis is restricted to altitudes between 100 and 350 km because the very low plasma densities at higher modeled altitudes can produce less reliable ion velocity estimates.

In the DR, large-amplitude oscillations in the meridional neutral winds coincide with fluctuations in the vertical ion drifts and electron density. During intervals of predominantly southward (equatorward) meridional winds, plasma is transported upward along the inclined geomagnetic field lines, producing localized electron density enhancements by raising the F-region plasma to altitudes where recombination rates are reduced. Conversely, northward (poleward) meridional winds are associated with downward plasma transport, resulting in localized electron density depletion. These relationships indicate that the storm-time TIDs are strongly influenced by wave-like perturbations in the meridional neutral wind field.

The vertical neutral winds exhibit temporal variations that closely follow the vertical ion drifts, which are driven by the thermospheric meridional winds. While it has been shown that vertical neutral winds can be driven by horizontal wind convergence (Anderson et al., 2012a, 2012b, 2012c; Cooper et al., 2009), few studies have explored the role of vertical ion drag on the vertical winds. Figure 12 shows an example in which there are waves of oscillating meridional winds, which result in both convergence/divergence areas, as well as regions of upward/downward ion drag. While it is beyond the scope of the current study, it is worth noting that this is an interesting interval in which the vertical momentum equation could be studied in detail to understand what sources are responsible for the observed ±10–30 m $s^{-1}$ mid-latitude vertical winds.

In the MH region, similar wind-driven behavior is present but with weaker variability. The meridional winds are less equatorward, resulting in reduced upward ion motion and consequently smaller density perturbations. However, significant longitudinal variability is present. For example, wave structures at roughly 254° longitude are not observed at 288°. This variability demonstrates that individual small-scale wind-driven features may not be consistently captured across locations, which complicates direct observational validation. The reduced variability observed at MH compared to the DR reflects local modulation of thermospheric forcing.

Together, Figures 11 and 12 demonstrate that the modeled ionospheric density structuring is associated with meridional wind variations that are highly structured in both time and longitude. GITM not only captures the large-scale storm-time ionospheric response but also resolves TIDs associated with wave-like variations in neutral winds and plasma drifts. The meridional wind disturbances are most likely generated by auroral and Joule heating at high latitudes. The spatial and temporal structure of this heating is strongly controlled by the specified electric potential and auroral patterns, represented here by the empirical models of Weimer (2005) and FTA (Wu et al., 2021). We note that the use of empirical models could lead to improper localized heating, which may drive the global structure to be relatively correct and some localized structures to be correct, such as in the trough region, but may produce improper localized heating that could drive inaccurate winds and dynamics (Deng & Ridley, 2007; Verkhoglyadova et al., 2016; Y. Yu et al., 2018; Hsu & Pedatella, 2023).

Finally, the vertical neutral winds in GITM show features similar to those in the vertical ion drifts. This correspondence suggests that some midlatitude vertical neutral winds may be influenced by meridional neutral winds that transport plasma along inclined magnetic field lines, followed by momentum transfer from the ions to the neutral atmosphere through vertical ion drag.

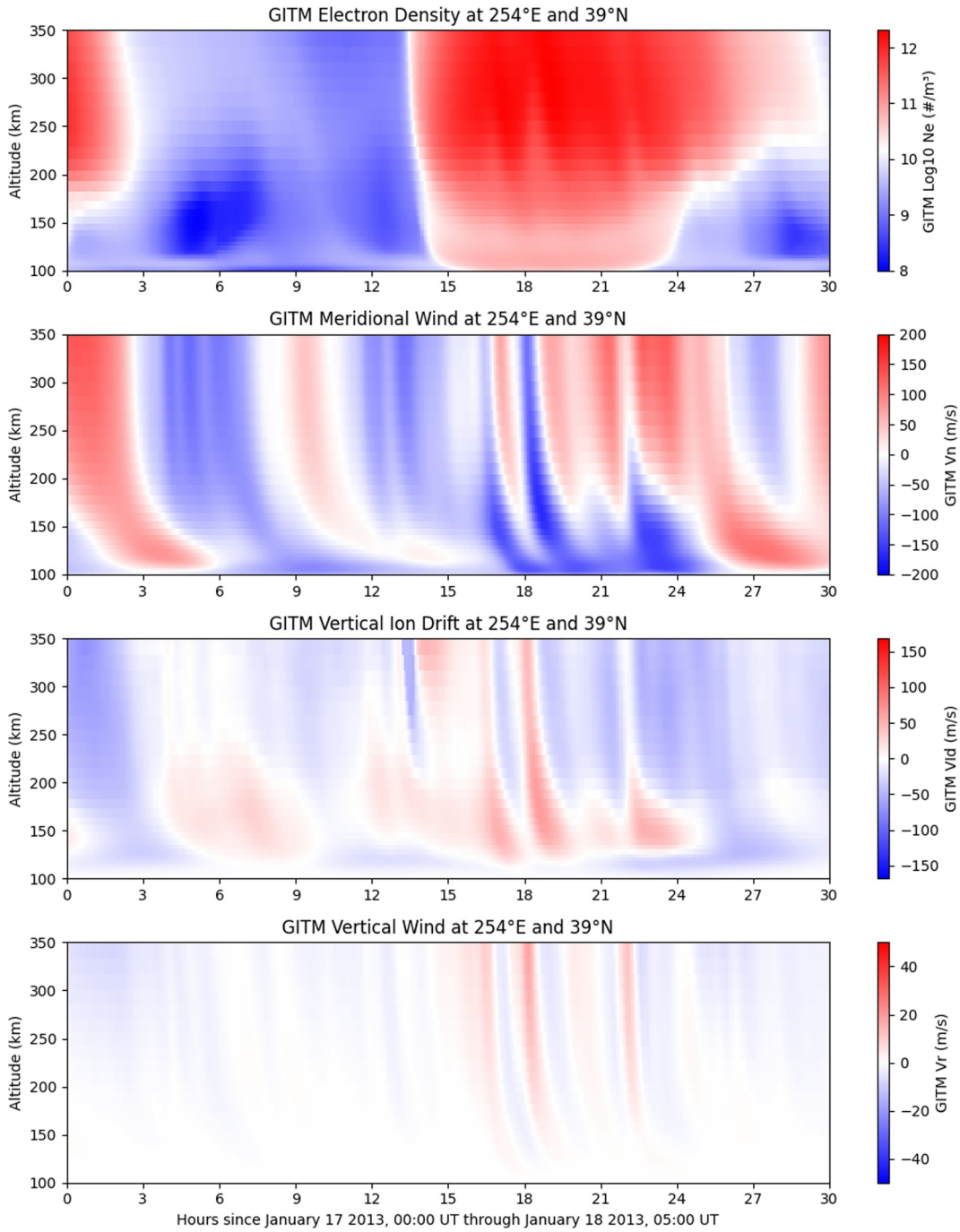



**Figure 11.** GITM-simulated electron density, meridional neutral wind, vertical ion drift, and vertical neutral wind within the depletion region (254°E, 39°N) from 00 UT on 17 January to 05 UT on 18 January 2013.

## 4. Conclusions

This article examined the weak geomagnetic storm of January 17 to 18, 2013, using a dual framework that integrates a first principles model, GITM, with the observation-driven TIDAS reconstruction and evaluates both against MH ISR measurements and GNSS-derived keograms. This study focused on improving the understanding of how storm-time coupling among neutral winds, plasma drifts, and thermospheric composition governs the development of both large-scale and mesoscale ionospheric structures. The analysis reveals a two-phase large-scale midlatitude response. During the first phase, beginning near 16:00 UT, equatorward meridional winds



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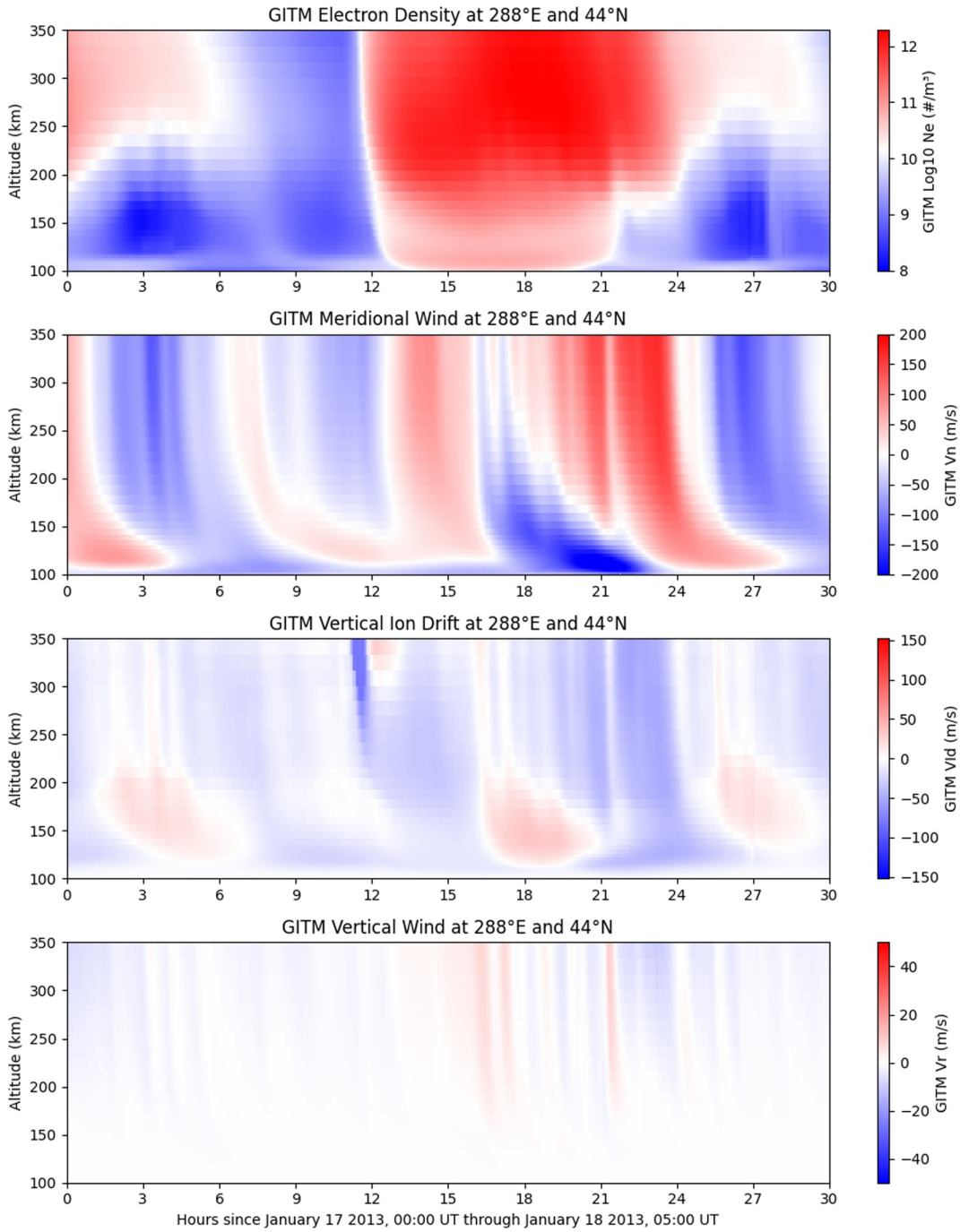



**Figure 12.** GITM-simulated electron density, meridional neutral wind, vertical ion drift, and vertical neutral wind at the Millstone Hill location (288°E, 44°N) from 00 UT on 17 January to 05 UT on 18 January 2013.

produce upward ion motion and elevate $hmF_2$ within the developing SED region. This wind-driven uplift supports the formation and equatorward progression of the SED plume, which remains evident through approximately 20:10 UT. During the second phase, beginning near 22:00 UT, a broad decrease in electron density occurs concurrently with the regional reduction and equatorward progression of thermospheric $O/N_2$ and weak downward ion motion. Localized density structures and mesoscale TIDs associated with additional neutral wind variability are superimposed on this large-scale evolution. The principal conclusions are as follows:



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1. Overall, GITM showed broad daytime agreement with TIDAS and reproduced the two-phase large-scale storm evolution. During the initial enhancement phase, the spatial correspondence among equatorward meridional winds, upward ion motion, and elevated $hmF_2$ indicates that meridional-wind-driven uplift is the primary mechanism associated with the formation and equatorward progression of the SED plume. GITM showed weaker agreement with TIDAS at night because it lacks plasmaspheric refilling mechanisms that resupply the midlatitude F region.
2. The later broad decrease in electron density occurred concurrently with a regional reduction and equatorward progression of thermospheric $O/N_2$ and downward ion motion. Reduced $O/N_2$ enhances plasma loss through molecular recombination, while downward ion motion transports plasma toward lower altitudes where recombination is faster. Together, these processes contribute to the large-scale depletion phase. Superimposed on this broad depletion, GITM produced localized and mesoscale electron-density structures associated with variations in meridional winds and vertical ion motion. These localized structures were more pronounced in GITM than in TIDAS and varied substantially with longitude.
3. Even though TIDAS assimilates MH ISR data, it does not reproduce the descending wave-like structures and localized variability that is seen consistently in both the ISR profiles and the GITM simulations. This indicates that the current spatial and temporal smoothing in TIDAS suppresses mesoscale TIDs and small-scale storm time structure, limiting its ability to represent wave-driven variability even in regions with good data coverage.
4. Within the localized DR and at MH, GITM showed spatial and temporal correspondence among meridional neutral winds, vertical ion drifts, and vertical neutral winds. This raises the possibility that the meridional winds could drive mid-latitude vertical winds through the ions being forced along slanted magnetic field lines. This avenue forms a recommended future exploration study.

This study identifies specific model and assimilative biases to target in future work, including nightside refilling in GITM and assimilation settings that preserve neutral dynamics imprints. Future work is recommended to explore adjustments to TIDAS spatial and temporal filtering that might be able to reveal some smaller-scale variations.

## Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

## Availability Statement

The Global Ionosphere Thermosphere Model (GITM) used to produce the simulations in this study is freely available at https://github.com/GITMCode/GITM. The TIDAS data assimilation results are available through Zenodo (Aa, Erickson, et al., 2022). Geomagnetic activity parameters were obtained from the OMNI data set maintained by the NASA Space Physics Data Facility and accessed through the OMNIWeb interface (NASA Space Physics Data Facility, 2026). GNSS TEC data and Millstone Hill incoherent scatter radar measurements were obtained from the Madrigal distributed data system (MIT Haystack Observatory, 2026). The TIMED/GUVI Level 3 $O/N_2$ data were obtained from the Johns Hopkins University Applied Physics Laboratory GUVI data portal (Johns Hopkins University Applied Physics Laboratory). Data for ionospheric TEC processing is provided from the following organizations: UNAVCO, Scripps Orbit and Permanent Array Center, Institut Geographique National, France, International GNSS Service, The Crustal Dynamics Data Information System (CDDIS), National Geodetic Survey, Instituto Brasileiro de Geografia e Estatística, RAMSAC CORS of Instituto Geográfico Nacional de la República Argentina, Arecibo Observatory, Low-Latitude Ionospheric Sensor Network (LISN), Canadian High Arctic Ionospheric Network, Institute of Geology and Geophysics, Chinese Academy of Sciences, China Meteorology Administration, Centro di Ricerche Sismologiche, Système d'Observation du Niveau des Eaux Littorales (SONEL), RENAG: REseau NAtional GNSS permanent (permanent, 2017), GeoNet - the official source of geological hazard information for New Zealand, Finnish Meteorological Institute, SWEPOS - Sweden, Hartebeesthoek Radio Astronomy Observatory, TrigNet Web Application, South Africa, Australian Space Weather Services, RETE INTEGRATA NAZIONALE GPS, Estonian Land Board, TU Delft, Western Canada Deformation Array, EUREF Permanent GNSS Network, GeoDAF: Geodetic Data Archiving Facility, African Geodetic Reference Frame (AFREF), Kartverket - Norwegian Mapping Authority, Geoscience Australia, IGS Data Center of Wuhan University, Pacific Northwest Geodetic Array, Nevada Geodetic Laboratory, Earth

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Observatory of Singapore, National Time and Frequency Standard Laboratory - Taiwan, and Korea Astronomy and Space Science Institute.

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